A Comprehensive Evaluation of Machine Learning Models for Skincare Ingredient Classification with Class Sparsity Reduction

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Abstract *:* The primary aim of this research was to evaluate machine learning models for classifying cosmetic ingredient functions based on ingredient names and descriptions.The COSING dataset was used as the primary data source, and TF-IDF feature extraction was applied to convert textual information into numerical representations suitable for classification. Three progressive stages were investigated: baseline model comparison using Logistic Regression, Decision Tree, Random Forest, and XGBoost; skincare-specific dataset refinement; and a proposed Class Sparsity Reduction Strategy that grouped similar skincare functions into three broader categories: Hydration & Conditioning, Environmental Protection, and Active Treatment. Experimental results showed that Random Forest achieved the strongest baseline performance. The skincare-specific dataset significantly improved classification accuracy, while the final class sparsity reduction framework achieved the best overall performance with 78.6% accuracy, 77.2% precision, 78.6% recall, and 73.2% F1-score, compared with 54.3% on the original dataset. These findings demonstrate that combining machine learning with domain-specific dataset refinement and semantic label restructuring can effectively improve automatic cosmetic ingredient analysis and intelligent skincare applications.

Keywords: *Cosmetic Ingredient Classification, Skincare Intelligence, Machine Learning, Random Forest, TF-IDF, COSING Dataset, Text Classification.*

GitHublink: https://github.com/rishindra-mateti-tech/Brainstrom.exe-HACKATHON-

1. INTRODUCTION

The cosmetic and skincare industry plays an important role in everyday life, as millions of people use products for personal care, skin health, and beauty enhancement. Modern cosmetic products contain a wide variety of ingredients, each designed to perform specific functions such as skin conditioning, moisturizing, cleansing, antioxidant protection, UV filtering, preservation, and fragrance enhancement. Understanding the functional role of these ingredients is valuable for manufacturers, researchers, dermatologists, regulatory agencies, and consumers. However, manually identifying ingredient functions from large ingredient databases and product labels is time-consuming, inconsistent, and difficult to scale [2].

Recent advances in artificial intelligence together with machine learning have developed efficient methods for automating these classification tasks. Machine learning models can identify patterns from historical ingredient data and generate accurate predictions for unseen samples. Since cosmetic ingredient names and descriptions are primarily text-based, text classification techniques are well suited for this problem. By converting textual information into numerical representations, ML-based algorithms are able to be used for predicting ingredient functions automatically [6].

This study uses the COSING (Cosmetic Ingredient) database provided by the European Commission as the primary dataset [1]. The COSING dataset was selected because it is one of the most reliable public cosmetic ingredient resources and contains real-world structured information such as ingredient names, descriptions, and official functional categories. Since the proposed system focuses on skincare ingredient classification, a dataset containing authentic ingredient labels and descriptions was essential. With more than 30,000 ingredient records, COSING is highly suitable for machine learning experiments involving multiclass classification, feature engineering, and performance benchmarking [1].

An Exploratory Data Analysis (EDA) of the dataset revealed notable class imbalance and variation in label distribution. Certain categories such as Skin Conditioning, Perfuming, and Cleansing included a high number of records, whereas categories like Anti-Seborrheic, Moisturising, and Skin Protecting had relatively fewer samples. This imbalance indicated that direct multiclass prediction would be difficult and highlighted the need for domain-based filtering and label reorganization. Visualizations such as class distribution graphs, confusion matrices, ROC curves, and performance comparison charts were used throughout the study to examine model behavior and validate results [14].

In this research, three progressive approaches were applied for cosmetic ingredient classification. The first approach compared several machine learning models, including Logistic Regression, Decision Tree, Random Forest, and XGBoost, using the complete cosmetic dataset. The second approach concentrated on developing a skincare-specific model by selecting only relevant skincare categories such as Skin Conditioning, Antioxidant, Humectant, Moisturising, and UV Absorber. The third approach introduced a Class Sparsity Reduction Strategy, in which similar skincare functions were merged into broader categories to reduce imbalance and improve model learnability [15].

TF-IDF feature extraction was applied to ingredient names and descriptions to convert text data into numerical vectors appropriate for machine learning methods [6]. The main purpose of this study was to evaluate and compare the results of different machine learning algorithms and assess whether domain-specific filtering and class restructuring improve prediction accuracy. This research can support skincare recommendation systems, ingredient labeling tools, cosmetic product analysis platforms, regulatory analytics, and consumer awareness applications, while also contributing to future AI-based skincare intelligence systems [3].

2. METHODOLOGY

2.1 System Architecture

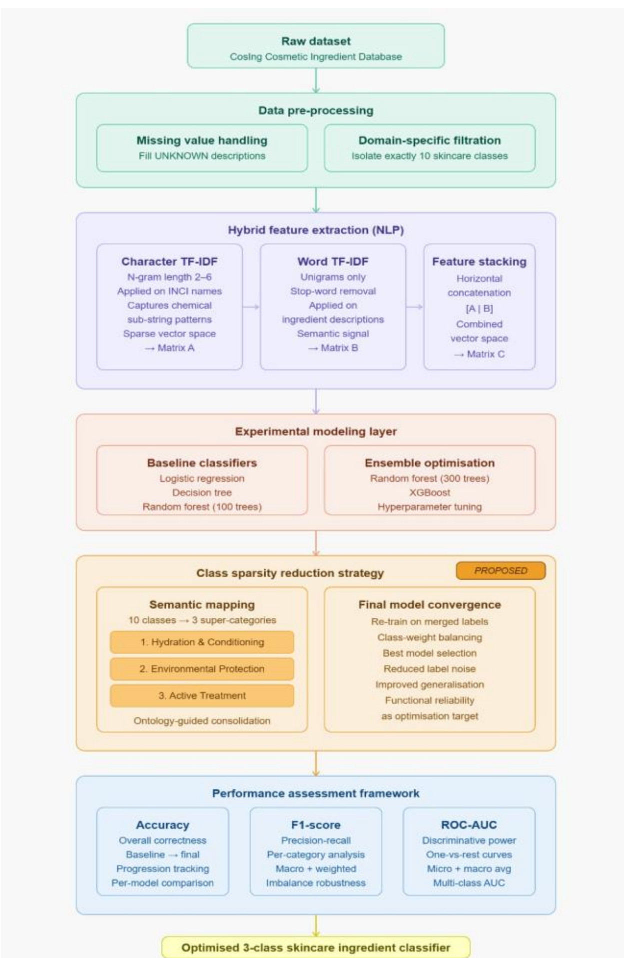

Figure 1: System architecture for proposed intelligent cosmetic ingredient classification framework.

Figure 1 presents an overall design of the proposed framework for intelligent cosmetic ingredient classification. The system follows an organized multi-stage workflow made up of dataset collection, data cleaning, attribute creation, machine learning classification, skincare-specific refinement, class sparsity reduction, and final model evaluation. The architecture was designed to progressively improve predictive performance through both algorithmic comparison and data-centric optimization.

2.2 Dataset Description

This project used the COSING (Cosmetic Ingredient) database provided by the European Commission as the primary dataset for cosmetic ingredient classification [1]. COSING is a trusted regulatory database containing detailed information about cosmetic ingredients, including ingredient names, functional uses, chemical descriptions, regulatory status, and identification details. The dataset contains more than 30,000 ingredient records distributed across 10 columns.

For this study, the most relevant fields selected were INCI Name, Function, and Chemical Description, as these columns contained the most useful and complete information for text-based classification. Columns with a high proportion of missing values, such as INN Name and Ph. Eur. Name, were excluded from the analysis.

Many ingredients in the dataset were linked to several listed functions. To make the classification process easier, only the primary listed function of each ingredient was selected as the target label. Common functional categories included Skin Conditioning, Perfuming, Antioxidant, Hair Conditioning, and Humectant.

The COSING dataset was used as the main source for training and testing ML models to automatically predict cosmetic ingredient functions based on ingredient names and descriptions. Figure 2 illustrates the top 25 most common primary functional categories in the dataset. It can be observed that Skin Conditioning is the dominant category, followed by Perfuming and Antioxidant, indicating clear class imbalance across labels. This imbalance motivated the later use of skincare filtering and class sparsity reduction strategies to improve model performance [14], [15].

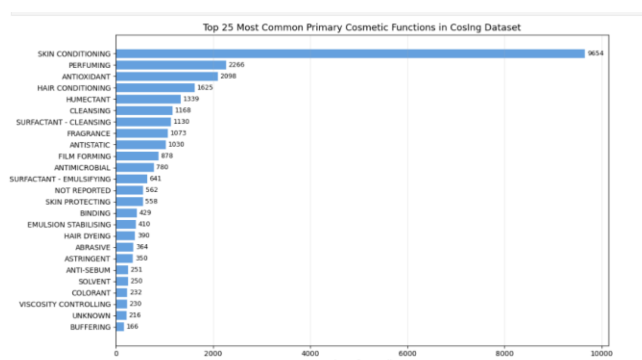

Figure 2: Top 25 Most Common Primary Cosmetic Functions in the COSING Dataset

2.3 System Process

In this project, a structured multi-stage process was followed to build an intelligent cosmetic ingredient classification system. The first stage involved collecting and preprocessing the COSING dataset, which contains more than 30,000 cosmetic ingredient records with names, descriptions, and functional categories [1]. Data cleaning included removing missing values, standardizing ingredient names, selecting the primary function of each ingredient, and eliminating rare classes with fewer than 10 samples. After preprocessing, the final dataset contained 30,020 samples across 59 functional categories.

During the second stage, attribute engineering was carried out with TF-IDF vectorization. Character-based TF-IDF was used for ingredient names to capture chemical naming structures, while word-based TF-IDF was used for ingredient descriptions to extract functional terms. These feature sets were combined into a sparse matrix and provided as input to ML algorithms [6].

In the third stage, multiple supervised models were tested, such as Logistic Regression, Decision Tree, Random Forest, and XGBoost, and were trained and evaluated using a stratified split with 80% training data and 20% testing data. Comparative experiments were conducted to identify the most effective baseline classifier [7], [8].

In the fourth stage, a skincare-specific dataset was created by selecting only skincare-related categories such as Skin Conditioning, Antioxidant, Humectant, Moisturising, UV Absorber, and related functions. This step reduced irrelevant cosmetic categories and focused the classification task on skincare ingredients.

In the final stage, a Class Sparsity Reduction Strategy was introduced by grouping similar skincare labels into three super-categories: Hydration & Conditioning, Environmental Protection, and Active Treatment. This restructuring reduced class imbalance and simplified the multiclass prediction task [14], [15]. The final framework was assessed using Accuracy, Precision, Recall, F1-score, confusion matrix analysis, along with ROC-AUC measures [11].

The overall methodology was designed to systematically compare baseline models, domain-specific filtering, and semantic label restructuring for intelligent cosmetic ingredient classification.

2.4 Data Preprocessing

The COSING dataset was first loaded into Python using the Pandas library for data handling and preprocessing [9]. Column names were standardized by removing extra spaces and converting them to uppercase format to maintain consistency across the dataset. Rows containing missing ingredient names or invalid target labels were removed, as they were not suitable for model training.

Missing values in the function column were handled wherever required. Functional labels with similar meanings, duplicate spellings, or minor variations were merged into common categories to reduce redundancy and improve label consistency. Since many ingredients had multiple listed functions, only the primary function was selected as the target label, converting the problem into a multiclass classification task.

To improve model reliability and reduce the impact of extreme class imbalance, rare categories containing fewer than 10 samples were removed from the dataset. Handling class imbalance is an important preprocessing step in multiclass learning problems [14]. After preprocessing, the final cleaned dataset contained 30,020 samples across 59 functional categories.

2.5 Feature Engineering

Since the COSING dataset primarily contains ingredient names and chemical descriptions in textual format, feature engineering was required to convert raw text into structured numerical representations appropriate for machine learning models. TF-IDF (Term Frequency–Inverse Document Frequency) was chosen since it is an widely used as well as effective technique used in text classification that highlights informative words and character patterns and lowers the impact of common words [5], [6].

For this study, character-level TF-IDF methods were applied on ingredient names in order to capture chemical naming structures, prefixes, suffixes, abbreviations, and other lexical patterns commonly found in cosmetic ingredients. This was particularly useful because many ingredient names contain complex scientific and technical terminology.

In addition, word-level TF-IDF was applied to ingredient descriptions to extract meaningful functional keywords such as conditioning, cleansing, antioxidant, moisturizing, and protecting. These textual cues helped the models better understand the functional role of each ingredient.

Finally, both feature representations were merged into one sparse input matrix, enabling the machine learning models to learn jointly from ingredient names and descriptions. This hybrid representation improved overall classification performance by leveraging both structural and semantic information [6].

2.6 Models Used

Different supervised machine learning techniques were trained and evaluated in this study to determine the best performing classifier for cosmetic ingredient function prediction. The selected algorithms represent different learning strategies, including linear, rule-based, ensemble, and boosting approaches.

Logistic Regression was selected as an initial benchmark since it is widely used for text classification tasks and performs efficiently on sparse TF-IDF feature spaces [6]. It was used to evaluate how effectively a linear model could classify ingredient functions in the COSING dataset.

Decision Tree was included as a rule-based classifier that creates hierarchical decision branches using feature values. This model was used to assess interpretable tree-based learning performance on cosmetic ingredient data, although it is generally more sensitive to overfitting.

Random Forest was selected as a tree-based ensemble model which uses many decision trees to increase prediction performance and generalization [8]. This model was evaluated in both baseline and optimized configurations. A baseline model with 100 trees was first tested, followed by an enhanced model with 300 trees and balanced class weighting to improve robustness on imbalanced data.

XGBoost was selected as a powerful gradient boosting method known for strong predictive performance, regularization capability, and computational efficiency [7]. It was used to compare boosting-based learning against bagging-based ensemble methods such as Random Forest.

Through this comparative framework, the study evaluated multiple ML strategies and identified a highly suitable approach for cosmetic ingredient prediction.

2.7 Hyperparameter Configuration

To maintain fair comparison and consistent model performance, specific hyperparameters were selected for each classifier. Logistic Regression was trained with a maximum of 500 iterations to ensure convergence on sparse TF-IDF features. Decision Tree used controlled maximum depth to reduce overfitting. Random Forest was evaluated in two configurations: a baseline model with 100 trees and an optimized model with 300 trees using balanced class weighting [8]. XGBoost was set up with 100 estimators, a learning rate of 0.1, and maximum tree depth of 6 [7]. These settings were chosen to achieve a balance between predictive accuracy and processing efficiency.

2.8 Model Training and Evaluation

After data preprocessing and feature engineering, the generated numerical feature matrix was assigned as the input variable (X), while the primary functional category of each ingredient was selected as the target label (y). The dataset was divided into training and testing sets using an 80:20 split, where 80% of the samples were used for model training and the remaining 20% were reserved for performance evaluation. Stratified sampling was applied to preserve the original class distribution in both subsets and ensure fair comparison.

Four supervised machine learning models were trained and evaluated: Logistic Regression, Decision Tree, Random Forest, and XGBoost. Each model was fitted using the training dataset and then tested on unseen samples from the testing dataset. Random Forest was further optimized by increasing the number of trees and applying balanced class weighting. Additional experiments were also conducted on the skincare-specific dataset and the final class sparsity reduction dataset.

Multiple evaluation metrics were used to measure classification performance. Accuracy was used to calculate the percentage of correctly classified samples. Precision measured the proportion of predicted labels that were correct, while Recall measured how many actual labels were successfully identified. The F1-score provided a balanced measure of Precision and Recall, particularly important for imbalanced multiclass data.

In addition, ROC-AUC analysis was used to evaluate class separability and predictive confidence, while confusion matrices were generated to examine class-wise prediction errors. These metrics provided a comprehensive comparison of all models and helped identify the most effective classification strategy.

3. EXRERIMENTAL SETUP

3.1 Experimental Environment

To validate how effective the proposed project framework was, multiple tests were carried out with the COSING cosmetic ingredient dataset under different classification settings. The experiments were implemented through Python with tools like Pandas, NumPy, Scikit-learn, SciPy, along with XGBoost. The main goal was to compare baseline machine learning models, evaluate skincare-specific classification, and measure the effectiveness of the proposed Class Sparsity Reduction Strategy.

3.2 Dataset Configurations

The first experiment was performed on the full preprocessed COSING dataset containing 30,020 samples distributed across 59 functional categories. The second experiment used a skincare-specific subset created by selecting 10 major skincare-related categories such as Skin Conditioning, Antioxidant, Humectant, Moisturising, UV Absorber, and related classes. The final experiment used a reduced three-class dataset created through semantic grouping of skincare functions into Hydration & Conditioning, Environmental Protection, and Active Treatment.

3.3 Train-Test Split Strategy

To achieve fair evaluation across all progressive experimental stages (baseline comparison, skincare-specific isolation, and class sparsity reduction), a consistent 80:20 stratified training-testing split was maintained. This specific ratio was chosen to provide sufficient data for training the complex ensemble algorithms while reserving a robust, unseen hold-out set for generating the final classification reports and ROC-AUC curves. A global random seed (random_state=42) was enforced across the entire Python environment to guarantee strict experimental reproducibility during hyperparameter tuning.

3.4 Feature Representation Pipeline

The computational pipeline utilized a dual-feature extraction setup using the Scikit-learn library. The TfidfVectorizer was instantiated twice to handle the different text structures. For ingredient names, it was configured at the character level with an n-gram range of 2 to 6 to mathematically capture chemical prefixes and suffixes. For ingredient descriptions, it was configured at the word level with standard English stop-word removal. The resulting matrices were horizontally concatenated into a unified sparse matrix. This exact feature space was held constant across all model evaluations to ensure that any variances in performance were strictly the result of algorithmic differences or dataset refinement.

3.5 Model Execution Configurations

During execution, the computational parameters for the evaluated algorithms were strictly controlled to balance performance with processing efficiency. Logistic Regression was executed with a ceiling of 500 iterations to accommodate the high dimensionality of the TF-IDF matrix. The ensemble models required specific parameter constraints; Random Forest was instantiated with parallel processing (n_jobs=-1) and class_weight=balanced_subsample to actively counteract the severe class imbalance during tree construction. XGBoost utilized the histogram-based tree method (tree_method=hist) to accelerate training on the sparse feature set, operating with a tree depth limit of 6, a learning rate set to 0.1, and a column subsampling ratio set at 0.3.

3.6 Evaluation Metrics

Model results were assessed through multiple metrics. Accuracy measured the percentage of samples classified correctly. Precision measured the correctness rate of predicted labels, while Recall measured the proportion of actual labels successfully identified. F1-score provided a balanced measure of Precision and Recall, particularly useful for imbalanced multiclass datasets. Confusion matrices were used for class-wise error analysis, and ROC-AUC curves were generated to evaluate class separability and predictive confidence.

3.7 Experimental Objective

This experimental setup enabled a fair and systematic comparison of baseline models, optimized ensemble methods, skincare-specific classification, and the final class sparsity reduction framework. The results were used to determine the best performing strategy for cosmetic ingredient prediction and intelligent skincare analysis.

4. EXPERIMENTAL RESULTS

The experimental evaluation provided important findings regarding the effectiveness of machine learning for cosmetic ingredient classification. Multiple supervised learning algorithms were trained with TF-IDF features extracted from ingredient names and descriptions. Results from Logistic Regression, Decision Tree, Random Forest, and XGBoost were compared across different experimental settings, including the full dataset, skincare-specific subset, and class sparsity reduction framework.

4.1 Baseline Model Performance

Initial tests were carried out on the full COSING dataset containing 59 functional categories. Three baseline models were evaluated: Logistic Regression, Decision Tree, and Random Forest (100 Trees). According to Table 1, Random Forest achieved the best baseline accuracy of 54.3%, followed by Logistic Regression at 52.5%, while Decision Tree showed the lowest performance with 44.9% accuracy.

In addition to accuracy, Random Forest also achieved the best Precision (51.8%), Recall (54.3%), and F1-score (49.1%) among the baseline models. Logistic Regression showed competitive performance with an F1-score of 46.3%, whereas Decision Tree produced the weakest overall results due to lower generalization ability.

These findings indicate that ensemble-based learning methods were more effective than single-tree and linear approaches for multiclass cosmetic ingredient classification using TF-IDF features. The comparative metric trends are illustrated in Figure 3, where Random Forest consistently performed better than the other baseline models in all evaluation metrics.

```
Training Logistic Regression... Accuracy = 0.5253
Training Decision Tree... Accuracy = 0.4494
Training Random Forest (100 trees)... Accuracy = 0.5433

======================================================================
Model                          Accuracy  Precision     Recall         F1
----------------------------------------------------------------------
Logistic Regression              0.5253     0.4745     0.5253     0.4629
Decision Tree                    0.4494     0.4174     0.4494     0.3939
Random Forest (100)              0.5433     0.5176     0.5433     0.4908
======================================================================
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Table 1: Baseline Model Performance Comparison on Full COSING Dataset

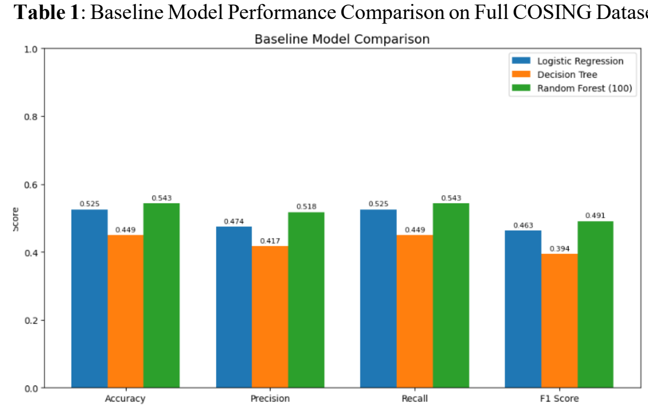

Figure 3: Baseline Model Comparison using Accuracy, Precision, Recall, and F1-score

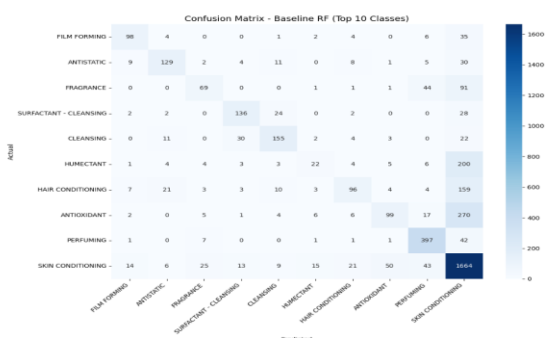

Figure 4: Confusion Matrix of Baseline Random Forest Model (Top 10 Functional Categories)

4.2 Optimized Full Dataset Performance

To improve performance, the Random Forest model was optimized by increasing the number of trees to 300 and applying balanced class weighting. The optimized model achieved 54.2% accuracy, while XGBoost reached 53.8% accuracy on the full dataset. Although improvements were limited, Random Forest remained the most stable and reliable model, suggesting that dataset complexity and class imbalance were major challenges.

4.3 Skincare-Specific Classification Results

A skincare-focused subset was created by selecting 10 major skincare-related categories such as Skin Conditioning, Antioxidant, Humectant, Moisturising, and UV Absorber. On this refined dataset, the optimized Random Forest model achieved the best overall accuracy of 69.7%, while XGBoost achieved 68.4%. This substantial improvement demonstrated that domain-specific filtering significantly enhanced prediction accuracy by removing unrelated cosmetic categories.

4.4 Class Sparsity Reduction Results

To further address class imbalance and semantic overlap, similar skincare labels were grouped into three broader super-categories: Hydration & Conditioning, Environmental Protection, and Active Treatment. This semantic restructuring reduced prediction complexity and created a more balanced classification framework. Using the optimized Random Forest model, the proposed framework achieved the highest overall accuracy of 78.6%, along with 77.2% Precision, 78.6% Recall, and 73.2% F1-score, representing a substantial improvement over both the original full dataset model and the skincare-specific model. As shown in Figure 6, the accuracy progression across all experimental stages demonstrates a clear upward trend, increasing from 54.3% in the baseline Random Forest model to 69.7% after skincare-specific filtering and finally to 78.6% after applying the Class Sparsity Reduction Strategy. A detailed comparison of Accuracy, Precision, Recall, and F1-score is presented in Figure 5, where the final three-class framework consistently outperformed the previous experimental settings across all evaluation metrics. Further analysis using the final confusion matrix in Figure 7 shows that the model performed exceptionally well on the Hydration & Conditioning category, achieving the highest number of correct predictions, while Environmental Protection and Active Treatment remained comparatively more challenging due to lower sample support and overlapping semantic characteristics. Overall, these findings confirm that semantic class restructuring greatly improved learnability, reduced label noise, and enhanced the overall effectiveness of cosmetic ingredient classification.

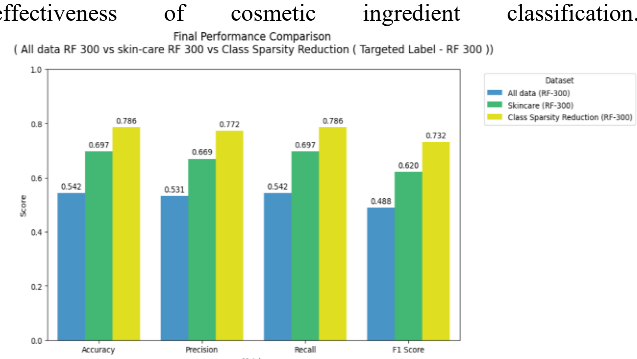

Figure 5: Final Performance Comparison of Full Dataset, Skincare Dataset, and Class Sparsity Reduction Framework

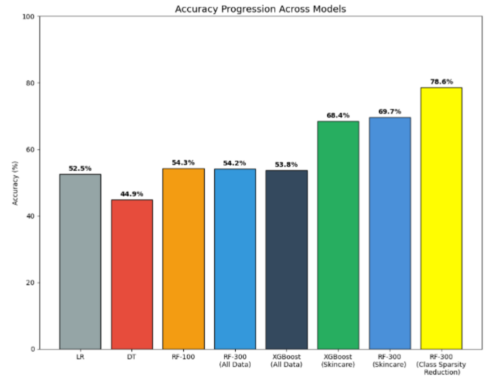

Figure 6: Accuracy Progression Across All Experimental Models

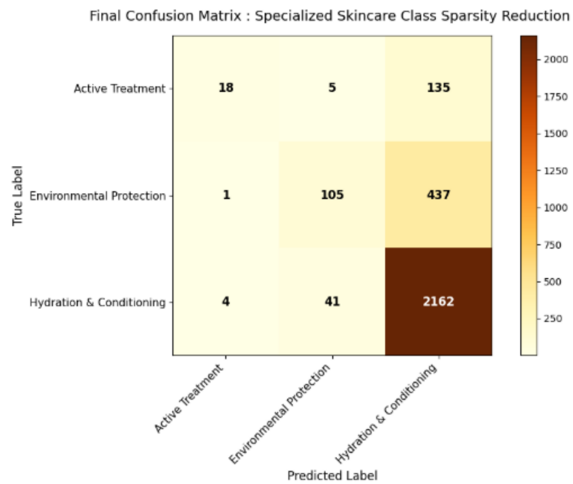

Figure 7: Final Confusion Matrix of Specialized Skincare Class Sparsity Reduction Model

5. SUMMARY OF EXPERIMENTAL RESULTS

a. Main Results

The findings indicate that Random Forest consistently outperformed the other machine learning approaches throughout this research. The baseline model achieved the best performance on the full dataset, while the skincare-specific model significantly improved classification accuracy. The highest performance was obtained using the proposed Class Sparsity Reduction Strategy, which increased accuracy to 78.6%. This demonstrates that domain-specific filtering and semantic label grouping were more effective than algorithm tuning alone.

Classifier	Accuracy
Logistic Regression	52.5%
Decision Tree	44.9%
Random Forest (100 Trees)	54.3%
Random Forest (300 Trees)	54.2%
XGBoost (Full Dataset)	53.8%
Skincare Random Forest	69.7%
Skincare XGBoost	68.4%
Class Sparsity Reduction Random Forest	78.6%

Table 2: Presents the complete overall results for all evaluated models across different experimental stages.

b. Ablation Study

An ablation study was conducted to measure how much each design stage contributed to the final system performance. The optimized Random Forest model achieved 54.2% accuracy on the original full cosmetic dataset. When the same model was trained on the skincare-specific subset, accuracy increased to 69.7%, demonstrating the strong positive impact of domain-focused data selection. After applying the proposed Class Sparsity Reduction Strategy, the final accuracy further increased to 78.6%, confirming that semantic label restructuring was the most influential improvement stage.

Similarly, XGBoost improved from 53.8% on the full dataset to 68.4% on the skincare dataset, showing that dataset refinement benefited multiple learning algorithms. As illustrated in Figure 8, ROC curve analysis showed stronger class separability in the final grouped framework compared to the original skincare model, indicating improved predictive confidence and reduced overlap among classes. In addition, Figure 9 presents the per-class F1-scores of the final framework, where Hydration & Conditioning achieved the highest score of 0.875, while Environmental Protection and Active Treatment achieved lower scores of 0.303 and 0.199, respectively.

These findings indicate that refining the dataset and reducing class sparsity had a greater influence on model performance than changing the classification algorithm alone, highlighting the importance of data-centric optimization in cosmetic ingredient classification.

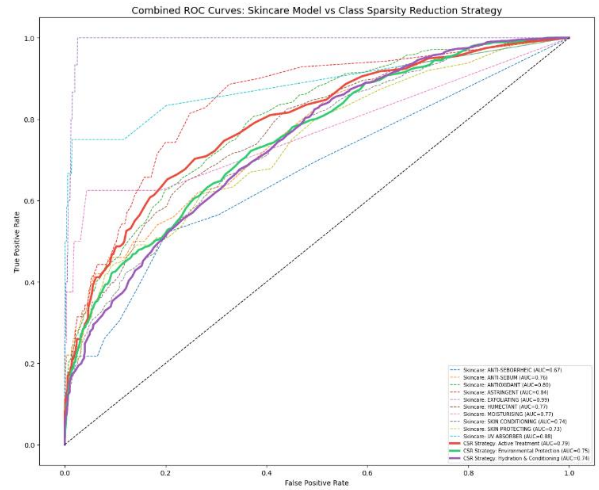

Figure 8: Combined ROC Curves: Skincare Model vs Class Sparsity Reduction Strategy

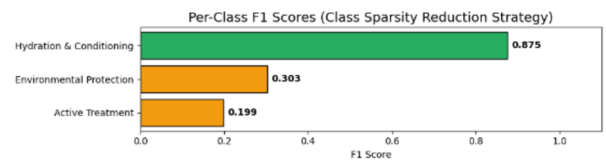

Figure 9: Per-Class F1 Scores for Final Class Sparsity Reduction Framework

c. Qualitative Analysis

Qualitative analysis was performed to understand model behavior across functional categories and compare class-wise performance before and after semantic label grouping. As shown in Figure 10, in the skincare-specific experiment, Skin Conditioning achieved the highest prediction performance with an F1-score of 0.81 and Recall of 0.97, mainly because it had the largest number of training samples and clearer textual patterns. Antioxidant and Humectant categories also showed comparatively stronger performance than other minority classes.

In contrast, categories such as Moisturising, Anti-Seborrheic, Skin Protecting, and UV Absorber produced lower prediction scores due to limited training samples, label sparsity, and semantic overlap with other skincare functions. This indicates that rare classes were more difficult for the model to learn effectively under the original multiclass setting.

After applying the final three-class Class Sparsity Reduction Strategy, overall class-wise performance improved substantially. The Hydration & Conditioning category achieved the strongest results with 0.88 F1-score and 0.98 Recall, while Environmental Protection showed moderate performance. Active Treatment remained the most challenging category with an F1-score of 0.20, likely due to fewer samples and more diverse functional characteristics.

Overall, the qualitative results confirm that frequent categories were learned more effectively, whereas sparse classes benefited significantly from semantic grouping and dataset restructuring.

Per-Class Performance (Baseline Skincare Model - RF-300):

	precision	recall	f1-score	support
ANTI-SEBORRHEIC	0.00	0.00	0.00	23
ANTI-SEBUM	0.80	0.08	0.15	50
ANTIOXIDANT	0.74	0.22	0.34	419
ASTRINGENT	0.36	0.06	0.10	70
EXFOLIATING	0.50	0.33	0.40	15
HUMECTANT	0.59	0.12	0.20	268
MOISTURISING	0.00	0.00	0.00	8
SKIN CONDITIONING	0.70	0.97	0.81	1931
SKIN PROTECTING	0.36	0.04	0.07	112
UV ABSORBER	1.00	0.25	0.40	12
accuracy			0.70	2908
macro avg	0.51	0.21	0.25	2908
weighted avg	0.67	0.70	0.62	2908

Per-Class Performance (Class Sparsity Reduction Strategy - RF-300):

	precision	recall	f1-score	support
Active Treatment	0.78	0.11	0.20	158
Environmental Protection	0.70	0.19	0.30	543
Hydration & Conditioning	0.79	0.98	0.88	2207
accuracy			0.79	2908
macro avg	0.76	0.43	0.46	2908
weighted avg	0.77	0.79	0.73	2908

Figure 10: Per-Class Classification Report Comparison: Skincare Model vs Final Class Sparsity Reduction Framework

6. ANALYSIS DISCUSSION

Random Forest performed better than Logistic Regression, Decision Tree, and XGBoost in all classification settings. Its ensemble learning structure was highly effective for handling sparse and high-dimensional TF-IDF text features while reducing overfitting. On the original full cosmetic dataset, the optimized Random Forest model achieved 54.2% accuracy. When the dataset was refined to skincare-specific categories, performance increased significantly to 69.7% accuracy. After applying the proposed Class Sparsity Reduction Strategy, the final model achieved the best overall performance with 78.6% accuracy, 77.2% precision, 78.6% recall, and 73.2% F1-score.

These findings indicate that data-centric improvements were more effective than algorithm-only optimization. Restricting the dataset to skincare-relevant categories enabled the model to learn more meaningful ingredient patterns, while grouping sparse and overlapping labels into broader functional categories reduced classification complexity. The final framework performed especially well on the Hydration & Conditioning category, which recorded the best per-class F1-score of 0.875, showing strong class separability and sufficient training support.

However, some categories remained more difficult to classify. Environmental Protection achieved an F1-score of 0.303, while Active Treatment achieved the lowest F1-score of 0.199. These lower scores were mainly caused by fewer training samples and functional overlap with other skincare categories. Many ingredients perform multiple roles, which naturally increases classification ambiguity.

The numerical results also show that model tuning alone produced only minor gains. XGBoost achieved 53.8% and optimized Random Forest achieved 54.2% on the full dataset. In contrast, dataset refinement and semantic label restructuring produced substantial improvements. This confirms that dataset quality, class balance, and label consistency can influence predictive performance more strongly than increasing model complexity.

7. TECHNICAL CHALLENGES AND FUTURE WORK

Multiple technical challenges arose during the development of this framework. The key challenge identified was severe label imbalance within the COSING dataset. Major categories such as Skin Conditioning contained many samples, while minority categories such as Moisturising, Anti-Seborrheic, and Skin Protecting had very limited data. This imbalance caused the model to favor majority classes and reduced prediction quality for rare categories.

Another challenge was semantic overlap among ingredient functions. Many cosmetic ingredients simultaneously act as moisturizers, antioxidants, protectants, or treatment agents. However, the dataset required a single primary label, which introduced ambiguity and label noise during training.

The project also relied mainly on textual features extracted from ingredient names and descriptions. It did not use chemical molecular structures, concentration percentages, formulation context, or user skin-type data. In addition, validation was performed primarily on the COSING dataset, so performance on commercial skincare product datasets remains untested.

With more time and resources, several improvements could be implemented. Advanced transformer-based models such as BERT, BioBERT, ChemBERTa, or LSTM could be used for richer semantic understanding. Class imbalance could be addressed using SMOTE, oversampling, or cost-sensitive learning. A multi-label classification framework could better represent ingredients with multiple simultaneous functions. Future work could also include multilingual support, real-time skincare recommendation systems, and deployment through web or mobile applications.

8. CONCLUSION

This study developed an intelligent cosmetic ingredient classification framework using machine learning and text analytics techniques. TF-IDF based feature engineering was applied to ingredient names and descriptions, and various supervised learning models were examined for function prediction. Among all baseline models, Random Forest achieved the strongest performance with 54.2% accuracy on the full dataset. A skincare-specific dataset improved classification performance to 69.7% accuracy, demonstrating the value of domain-focused learning.

The proposed Class Sparsity Reduction Strategy produced the best overall results, achieving 78.6% accuracy, 77.2% precision, 78.6% recall, and 73.2% F1-score. The Hydration & Conditioning category recorded the best per-class F1-score of 0.875, while Active Treatment remained the most challenging category with a low F1-score of 0.199. These findings clearly show that domain-specific filtering and semantic label restructuring can significantly improve cosmetic ingredient classification performance. Overall, the proposed framework provides a practical foundation for skincare recommendation systems, ingredient labeling tools, cosmetic product analysis platforms, and future AI-driven skincare intelligence applications.

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10.AUTHOR CONTRIBUTIONS

Rishindra Mateti: Conceived the end-to-end machine learning framework, designed the Class Sparsity Reduction (CSR) Strategy to resolve target label imbalance, implemented advanced NLP feature engineering using character-level TF-IDF vectorization, and executed optimized XGBoost training and fine-tuning to achieve high-performance classification for ingredient functionality.

Tejaswi Kancherla: Executed data preprocessing and feature standardization, implemented stratified train-test splitting to preserve class distributions, performed hyperparameter tuning for baseline models, and conducted ROC-AUC performance analysis to evaluate classification reliability across various functional categories.

Surya Thota: Performed multi-source data ingestion and Exploratory Data Analysis (EDA), established baseline benchmarks using Logistic Regression, and engineered the final results visualization suite including accuracy progression charts and confusion matrices in this project.